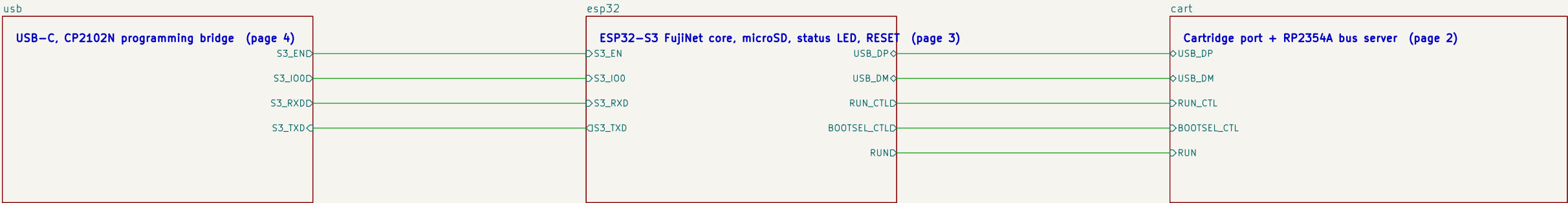


FujiNet for the Bally Astrocade – Rev0

One cassette–shell cartridge carrying both halves of the Astrocade FujiNet stack. Signal flow left to right:
PC / power → USB–C + CP2102N → ESP32–S3 (FujiNet, WiFi, microSD) → native USB → RP2354A → console bus.



File: usb-uart.kicad_sch

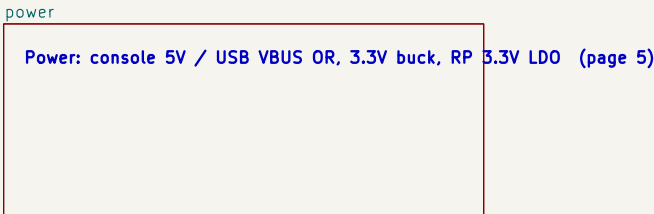
File: esp32s3-sd.kicad_sch

File: cart-rp2354a.kicad_sch

UART + auto–program: esptool resets the S3 into its ROM loader over DTR/RTS.

USB host of the RP (CDC device, VID 0xCafe): embeds the RP firmware and re–flashes it over PICOBOOT when it changes: IO4 pulses RUN while IO5 holds QSPI_SS low.

Serves the 8K cart window and the FujiNet mailbox straight off the 5 V bus (5 V–tolerant pads, no level shifters). Edge: A0–A12, D0–D7, /CCS (pre–decoded enable), +5V, 3 × GND. The console only ever reads the cart.



File: power.kicad_sch

Not verifiable in CAD (bring–up checklist, README):
– console cold start: +3V3_RP vs CA*/CCS_N rise (scope)
– /CCS–to–data timing against the 280 read cycle
– console 5 V headroom under WiFi bursts
– blade reach vs the land escape holes (17.4 mm)

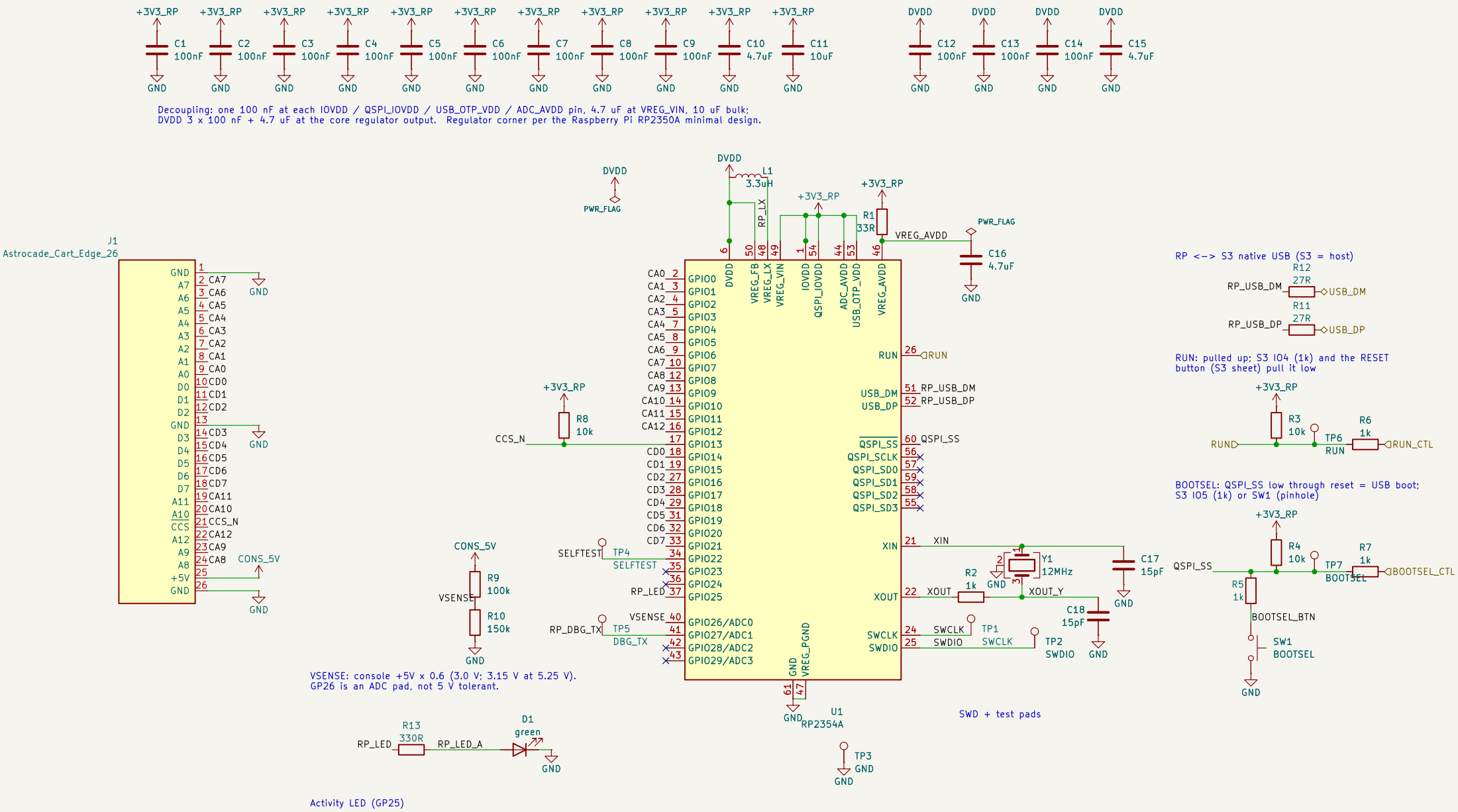
Rails (power symbols, global):
CONS_5V console +5V, edge land 25
VBUS USB–C
+5V CONS_5V / VBUS diode–OR → buck, LDO, WS2812
+3V3 AP63203 buck → ESP32–S3, CP2102N, microSD
+3V3_RP AP2112K LDO → RP2354A (all supplies)
DVDD RP2350 on–chip core SMPS (1.1 V)

License CERN–OHL–W–2.0. Circuit blocks from FujiNet–INTV–Rev0 (PINTY CARD) and the ESP32–S3–DevKitC–1; RP2350 regulator corner from the Raspberry Pi RP2350A minimal design (MIT).

Rev0 audit 2026–10–02: kicad–happy + manufacturer datasheets; see docs/design–review–rev0.md		
CERN–OHL–W–2.0 (derived from FujiNet–INTV–Rev0 / PINTY CARD)		
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py		
FujiNet		
Sheet: /		
File: FujiNet–Astrocade–Rev0.kicad_sch		
Title: FujiNet Astrocade Rev0		
Size: A3	Date: 2026–10–02	Rev: 0
KiCad E.D.A. 10.0.6	Id: 1/5	

CARTRIDGE PORT + RP2354A BUS SERVER

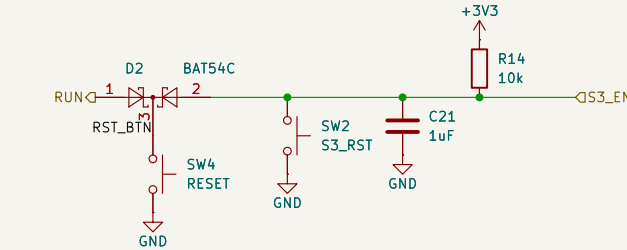
Bally Astrocade cassette port, 26 contacts on the PCB UNDERSIDE (the console blade presses up), Tilton 1–26.
The RP2354A (RP2350A + 2 MB flash in package) serves the 8K cart window straight off the 5 V bus: A0–A12 = GP0–12, /CCS = GP13, D0–D7 = GP14–21 (astrocade_cart.h). No buffers: the RP2350 fault-tolerant pads take 5.5 V while IOVDD is up, and +3V3_RP (power sheet, LDO) follows the console rail as it rises. The firmware drives D0–D7 only while /CCS is low AND VSENSE is high, so a USB-powered cart never drives a dead console.



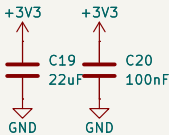
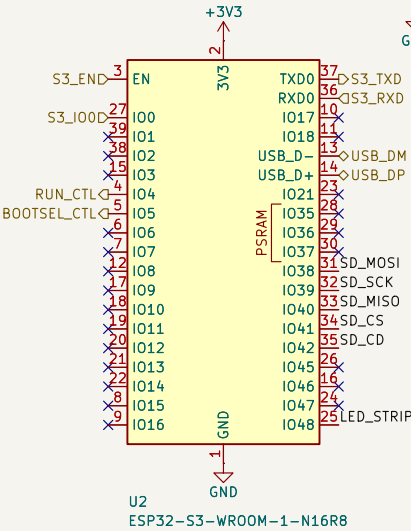
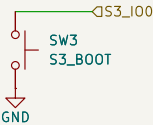
ESP32-S3 FUJINET CORE

ESP32-S3-WROOM-1-N16R8 runs fujinet-firmware (build fujiversal-astrocade, include/pinmap/fujiversal-astrocade.h). It is the USB HOST of the RP2354A (IO19/IO20, native USB) and re-flashes it over PICOB00T: IO4 -> RP RUN, IO5 -> RP QSPLSS (1k series each, on the RP sheet), driven LOW to assert, inputs when idle. IO26-IO37 belong to the module's flash/PSRAM (N16R8) and stay unconnected.

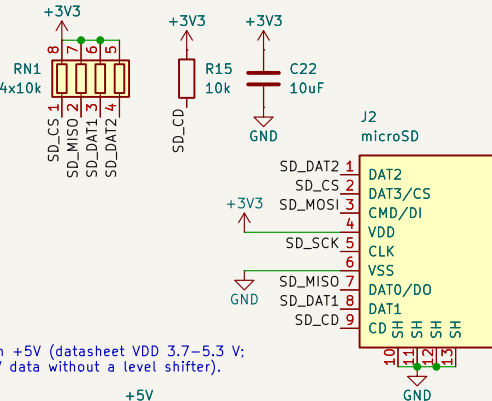
RESET (top face) pulls RP RUN and S3 EN low through a common-cathode BAT54C: neither reset line can drive the other. EN: 10k / 1 uF power-on delay.



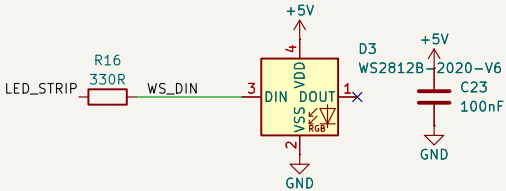
S3 BOOT (IO0)



microSD (SPI). CD closes to the shell with a card in: SD_CD reads LOW = card present.



Status LED: WS2812B-2020 on +5V (datasheet VDD 3.7-5.3 V; VIH 2.7 V takes the S3's 3.3 V data without a level shifter).



Rev0 audit 2026-10-02: kicad-happy + manufacturer datasheets; see docs/design-review-rev0.md CERN-OHL-W-2.0 (derived from FujiNet-INTV-Rev0 / PINTY CARD) Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /esp32/

File: esp32s3-sd.kicad_sch

Title: FujiNet Astrocade Rev0 - ESP32-S3 FujiNet core, microSD, status LED

Size: A4

Date: 2026-10-02

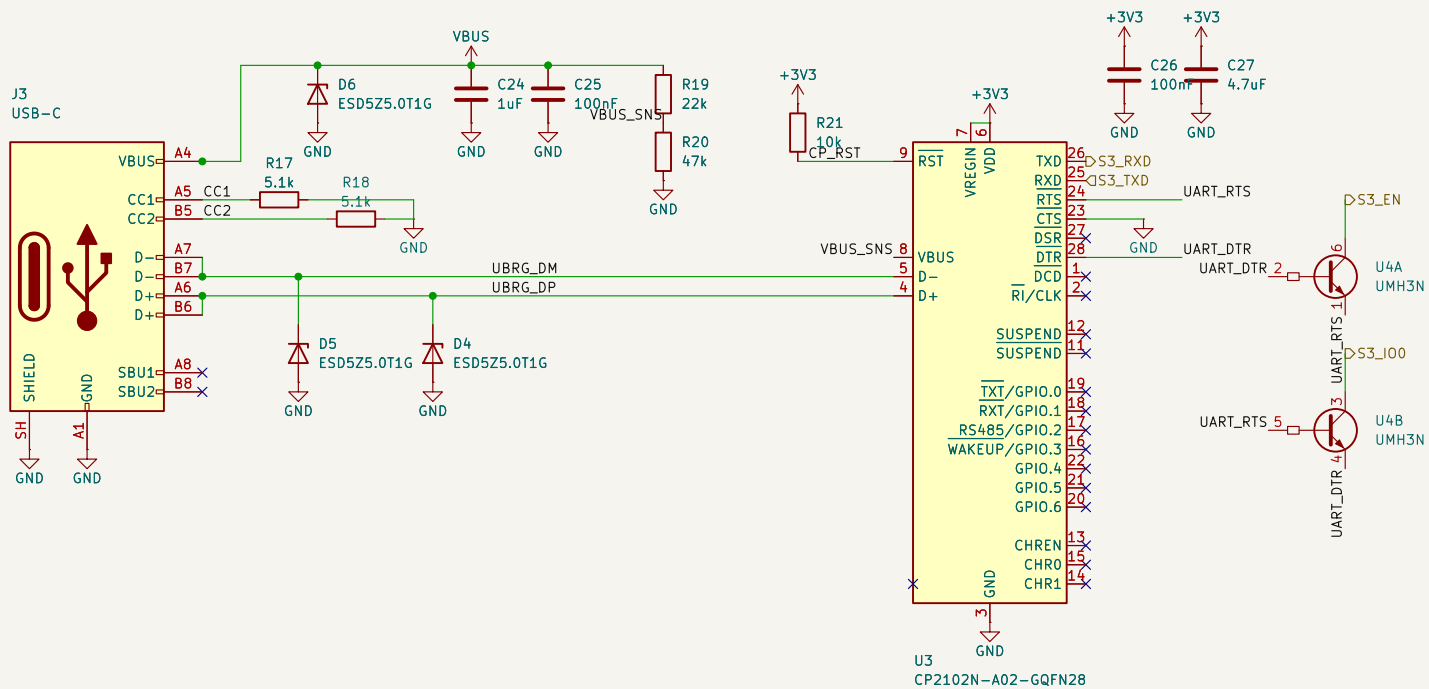
Rev: 0

KiCad E.D.A. 10.0.6

Id: 3/5

USB-C: POWER + ESP32-S3 PROGRAMMING BRIDGE

UFP (5.1k Rd on CC1/CC2). ESD5Z5.0 on D+, D-, VBUS at the connector. CP2102N is self-powered from +3V3 (VDD = VREGIN, regulator bypassed); its VBUS pin only senses the cable through 22k/47k (3.4 V at 5 V). UMH3N = the ESP32-S3-DevKitC-1 auto-program pair: esptool drives DTR/RTS to pulse EN and hold IO0.



Rev0 audit 2026-10-02: kicad-happy + manufacturer datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-INTV-Rev0 / PINTY CARD)
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py - edit those, not this file
FujiNet

Sheet: /usb/
File: usb-uart.kicad_sch

Title: FujiNet Astrocade Rev0 – USB-C, CP2102N programming bridge

Size: A4	Date: 2026-10-02
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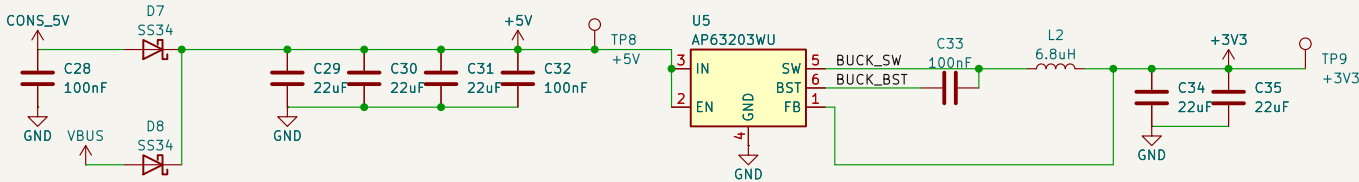
KiCad E.D.A. 10.0.6

Rev: 0

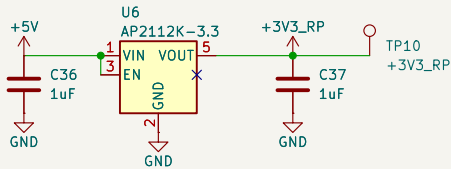
Id: 4/5

POWER

Console +5V (edge land 25, CONS_5V) and USB VBUS are diode-ORed into +5V: neither source can back-feed the other (an unpowered console, or the USB host). SS34 VF ~0.35 V at the cart's ~0.4 A, so +5V is 4.4–4.9 V; there is no 5 V logic on this board.



AP63203 3.3 V / 2 A buck: ESP32–S3 (WiFi bursts ~400 mA), CP2102N, microSD. 4 ms soft-start (tSS).



AP2112K–3.3 LDO -> +3V3_RP: the WHOLE RP2354A (IOVDD, QSPI/USB/ADC supplies, VREG_VIN + VREG_AVDD). In dropout it follows +5V as the console rail rises (1 V/ms into 21 uF = 21 mA, under the 50 mA fold-back limit), so the 5 V-tolerant pads are never unpowered with the console bus driving them (RP2350 datasheet: FT pads tolerate 5.5 V "provided IOVDD is powered to 3.3 V"). The buck instead waits for UVLO, then soft-starts over 4 ms. VREG_VIN and VREG_AVDD rise together (p.402).



PWR_FLAGS: the rails above are driven through diodes / inductors (passive pins).

Rev0 audit 2026–10–02: kicad-happy + manufacturer datasheets; see docs/design-review-rev0.md
CERN–OHL–W–2.0 (derived from FujiNet–INTV–Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py from tools/design.py + tools/sch_layout.py – edit those, not this file

FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet Astrocade Rev0 – Power: console 5V / USB VBUS OR, 3.3V buck

Size: A4	Date: 2026–10–02	Rev: 0
KiCad E.D.A. 10.0.6		Id: 5/5